QUESTION: Develop a menu-driven Java program called StudentUtilityProgram that demonstrates Java programming fundamentals. The program should accept student details and perform various operations using variables, data types, operators, input/outputs.

Steps for Execution

1.Open IntelliJ IDEA / Eclipse / VS Code.

2.Create a Java project.

3.Create a class named StudentUtilityProgram.

4.Enter the given code and save it.

5.Run the StudentUtilityProgram class.

6.Select Part 1, Part 2, Part 3, or Part 4.

7.Enter the required case number 1–4.

8.Enter the required input values.

9.The selected operation displays its result.

Algorithm

1.Start.

2.Create a Scanner object.

3.Display the main menu with 4 parts.

4.Read the user's choice.

5.Use switch to select the required part.

6.Part 1 – Student Operations

7.Select case 1–4.

8.Display student information, read student details, calculate total/percentage, or display the complete result.

9.Part 2 – Decision Making

10.Check even/odd.

11.find the largest of three numbers.

12.Determine grade from percentage.

13.Display the day using day number.

14.Part 3 – Loops

15.Generate multiplication table.

16.Display numbers from 1 to N.

17.Calculate sum from 1 to N.

18.Generate Fibonacci series.

19.Part 4 – Methods

20.Create a StudentUtilityProgram object.

21.Call the selected method:

factorial()

prime()

maximum()

areaOfCircle()

Display the result.

22.Stop.

UML Class Diagram

```
┌─────────────────────────────────────┐
|        StudentUtilityProgram        |
├─────────────────────────────────────┤
| + main(String[]) : void             |
| + factorial(Scanner) : void         |
| + prime(Scanner) : void             |
| + maximum(Scanner) : void           |
| + areaOfCircle(Scanner) : void      |
└─────────────────┬───────────────────┘
                  |
                  | uses
                  ▼
          ┌────────────────┐
          |    Scanner     |
```

├──────────────────────┤

| + nextInt() : int |

| + nextDouble() |

| + nextLine() |

└───────────────────────┘tatements, conditional statements, loops, and parameterized methods.

Code:

```
import java.util.Scanner;
public class StudentUtilityProgram {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("1. Part 1");
        System.out.println("2. Part 2");
        System.out.println("3. Part 3");
        System.out.println("4. Part 4");
        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();
        switch (choice) {
            case 1:
                System.out.print("Enter case 1-4 ");
                int achoice = sc.nextInt();
                String name = "";
                int rollno;
                double marks1, marks2, marks3, total, percentage;
                switch (achoice) {

                    case 1:
                        System.out.println("Student Information");
                        System.out.println("Name : priya");
                        System.out.println("Roll No : 183");
                        break;

                    case 2:
                        System.out.println("Enter Student Name: ");
                        name = sc.nextLine();

                        System.out.println("Enter Roll Number: ");
                        rollno = sc.nextInt();
```

```
    System.out.println("Enter English Marks: ");
    marks1 = sc.nextDouble();

    System.out.print("Enter Maths Marks: ");
    marks2 = sc.nextDouble();

    System.out.print("Enter Social Marks: ");
    marks3 = sc.nextDouble();
    break;

  case 3:
    System.out.print("Enter English Marks: ");
    marks1 = sc.nextDouble();

    System.out.print("Enter Maths Marks: ");
    marks2 = sc.nextDouble();

    System.out.print("Enter Social Marks: ");
    marks3 = sc.nextDouble();

    total = marks1 + marks2 + marks3;
    percentage = (total / 300) * 100;

    System.out.println("Total = " + total);
    System.out.println("Percentage = " + percentage);
    break;

  case 4:
    System.out.print("Enter Student Name: ");
    name = sc.nextLine();

    System.out.print("Enter Roll Number: ");
    rollno = sc.nextInt();

    System.out.print("Enter English Marks: ");
    marks1 = sc.nextDouble();

    System.out.print("Enter Maths Marks: ");
    marks2 = sc.nextDouble();
```

```
            System.out.print("Enter Social Marks: ");
            marks3 = sc.nextDouble();

            total = marks1 + marks2 + marks3;
            percentage = (total / 300) * 100;

            System.out.println("\n----- RESULT -----");
            System.out.println("Name : " + name);
            System.out.println("Roll No : " + rollno);
            System.out.println("Total : " + total);
            System.out.println("Percentage : " + percentage);
            break;

        default:
            System.out.println("Invalid Choice");
    }
    break;
case 2:
    System.out.println("enter case 1-4");
    int bchoice = sc.nextInt();
    switch (bchoice) {
        case 1:
            System.out.print("Enter a number: ");
            int n = sc.nextInt();

            if (n % 2 == 0)
                System.out.println("Even Number");
            else
                System.out.println("Odd Number");
            break;

        case 2:
            System.out.print("Enter first number: ");
            int a = sc.nextInt();

            System.out.print("Enter second number: ");
            int b = sc.nextInt();

            System.out.print("Enter third number: ");
            int c = sc.nextInt();
```

```
        if (a >= b && a >= c)
            System.out.println("Largest = " + a);
        else if (b >= a && b >= c)
            System.out.println("Largest = " + b);
        else
            System.out.println("Largest = " + c);
        break;

    case 3:
        System.out.print("Enter percentage: ");
        double per = sc.nextDouble();

        if (per >= 90)
            System.out.println("Grade A");
        else if (per >= 75)
            System.out.println("Grade B");
        else if (per >= 60)
            System.out.println("Grade C");
        else if (per >= 40)
            System.out.println("Grade D");
        else
            System.out.println("Fail");
        break;

    case 4:
        System.out.print("Enter day number (1-7): ");
        int day = sc.nextInt();

        switch (day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            case 4:
                System.out.println("Thursday");
                break;
```

```
            case 5:
                System.out.println("Friday");
                break;
            case 6:
                System.out.println("Saturday");
                break;
            case 7:
                System.out.println("Sunday");
                break;
            default:
                System.out.println("Invalid Day");
        }
        break;

    }
case 3:
    System.out.println("enter case 1-4");
    int cchoice = sc.nextInt();

    switch (cchoice) {

        case 1:
            System.out.print("Enter a number: ");
            int num = sc.nextInt();

            System.out.println("Multiplication Table:");
            for (int i = 1; i <= 10; i++) {
                System.out.println(num + " x " + i + " = " + (num * i));
            }
            break;

        case 2:
            System.out.print("Enter N: ");
            int n = sc.nextInt();

            System.out.println("Numbers from 1 to " + n + ":");
            for (int i = 1; i <= n; i++) {
                System.out.print(i + " ");
            }
            break;
```

```
        case 3:
            System.out.print("Enter N: ");
            int m = sc.nextInt();

            int sum = 0;
            for (int i = 1; i <= m; i++) {
                sum = sum + i;
            }

            System.out.println("Sum = " + sum);
            break;

        case 4:
            System.out.print("Enter number of terms: ");
            int terms = sc.nextInt();

            int a = 0, b = 1, c;

            System.out.println("Fibonacci Series:");
            for (int i = 1; i <= terms; i++) {
                System.out.print(a + " ");
                c = a + b;
                a = b;
                b = c;
            }
            break;


    }
case 4:
    System.out.println("Enter case 1-4");
    int dchoice = sc.nextInt();

    StudentUtilityProgram obj = new StudentUtilityProgram();

    if (dchoice == 1)
        obj.factorial(sc);
    else if (dchoice == 2)
        obj.prime(sc);
    else if (dchoice == 3)
        obj.maximum(sc);
```

```java
            else if (dchoice == 4)
                obj.areaOfCircle(sc);
            else
                System.out.println("Invalid Choice");

            break;

    }
}

void factorial(Scanner sc) {
    System.out.print("Enter a number: ");
    int n = sc.nextInt();

    int fact = 1;
    for (int i = 1; i <= n; i++) {
        fact = fact * i;
    }

    System.out.println("Factorial = " + fact);
}

void prime(Scanner sc) {
    System.out.print("Enter a number: ");
    int num = sc.nextInt();

    int count = 0;
    for (int i = 1; i <= num; i++) {
        if (num % i == 0)
            count++;
    }

    if (count == 2)
        System.out.println("Prime Number");
    else
        System.out.println("Not a Prime Number");
}

void maximum(Scanner sc) {
    System.out.print("Enter first number: ");
    int a = sc.nextInt();
```

```
        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        if (a > b)
            System.out.println("Maximum = " + a);
        else
            System.out.println("Maximum = " + b);
    }

    void areaOfCircle(Scanner sc) {
        System.out.print("Enter radius: ");
        double r = sc.nextDouble();

        double area = 3.14 * r * r;

        System.out.println("Area of Circle = " + area);
    }
    }
```

output:

```
1. Part 1
    2. Part 2
    3. Part 3
    4. Part 4
Enter your choice:

  PART 1

Enter case 1-4 1

Student Information
Name : priya
Roll No : 183


Enter case 1-4 2

Enter Student Name:
Enter Roll Number:
```

Enter English Marks:
Enter Maths Marks:
Enter Social Marks:

Enter case 1-4 3

Enter English Marks: 80
Enter Maths Marks: 90
Enter Social Marks: 85

Total = 255.0
Percentage = 85.0

Enter case 1-4 4

Enter Student Name: Priya
Enter Roll Number: 183
Enter English Marks: 80
Enter Maths Marks: 90
Enter Social Marks: 85

----- RESULT -----
Name : Priya
Roll No : 183
Total : 255.0
Percentage : 85.0

PART 2

enter case 1-4

Case 1:
Enter a number: 10
Even Number

Case 2:
Enter first number: 10

Enter second number: 20
Enter third number: 15
Largest = 20

Case 3:
Enter percentage: 85
Grade B

Case 4:
Enter day number (1-7): 3
Wednesday

PART 3

Case 1:
Enter a number: 5

Multiplication Table:

5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Case 2:
Enter N: 5

Numbers from 1 to 5:

1 2 3 4 5

Case 3:

Enter N: 5

Sum = 15

Case 4:
Enter number of terms: 7

Fibonacci Series:
0 1 1 2 3 5 8

PART 4

Case 1:
Enter a number: 5

Factorial = 120

Case 2:
Enter a number: 7

Prime Number

Case 3:
Enter first number: 25
Enter second number: 15

Maximum = 25

Case 4:
Enter radius: 5

Area of Circle = 78.5